

Chapter 2 Vacuum Science and Technology

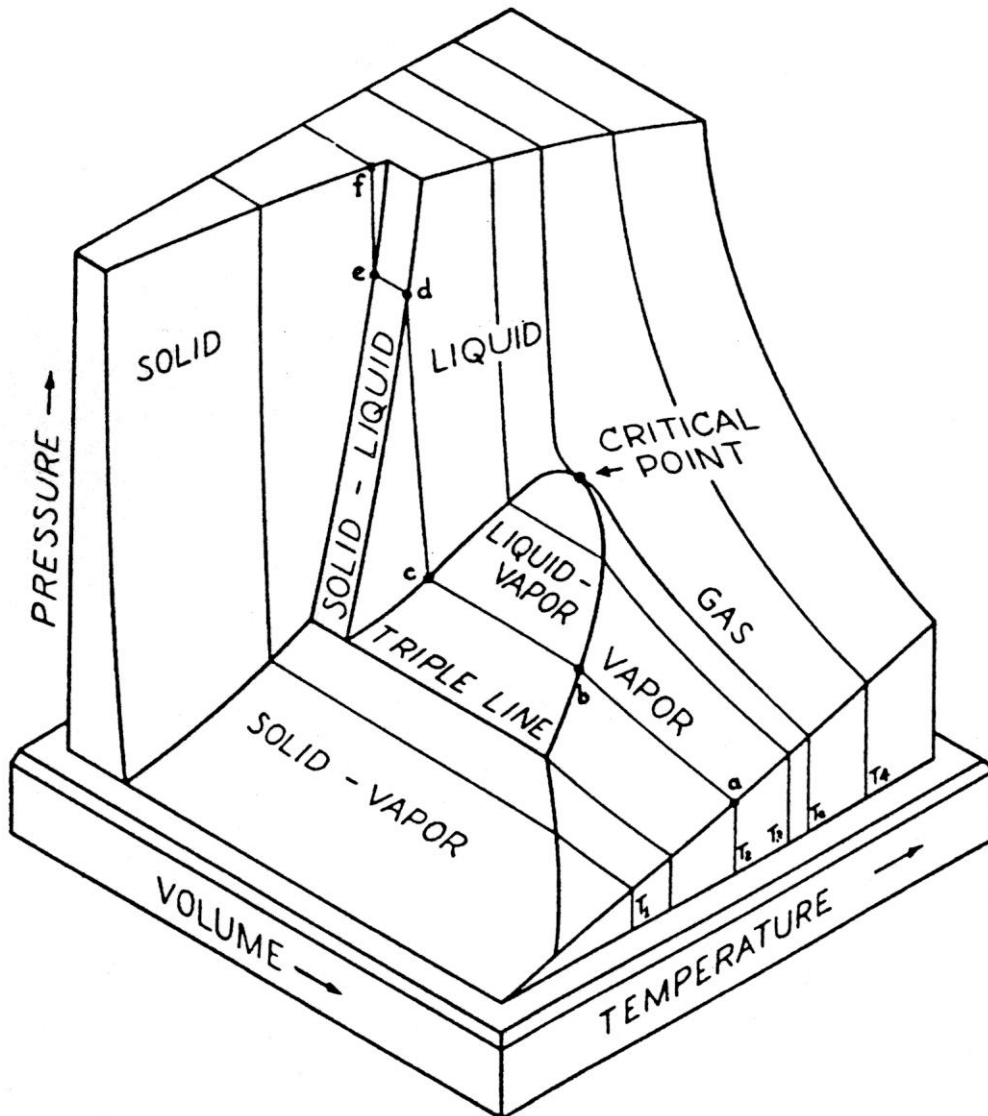
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2.1 Vapors and Gases

The behavior of vapors and gases on both the macroscopic and molecular scales is fundamental to an understanding of thin-film deposition from the vapor phase. We will examine the molecular interaction of gases and will relate that behavior to the macroscopic, measurable properties such as temperature and pressure.

The behaviors of **vapors** and **gases** need to be first distinguished. At a fixed temperature, T , a **vapor** can be condensed to a liquid or solid by compressing it sufficiently, whereas a **gas** cannot.

p - V - T phase diagram



Look at a p - V plane at T_2 . It intercepts the surface along a-b-c-d-e-f:

At a, it is a vapor phase.

At b, condensation starts.

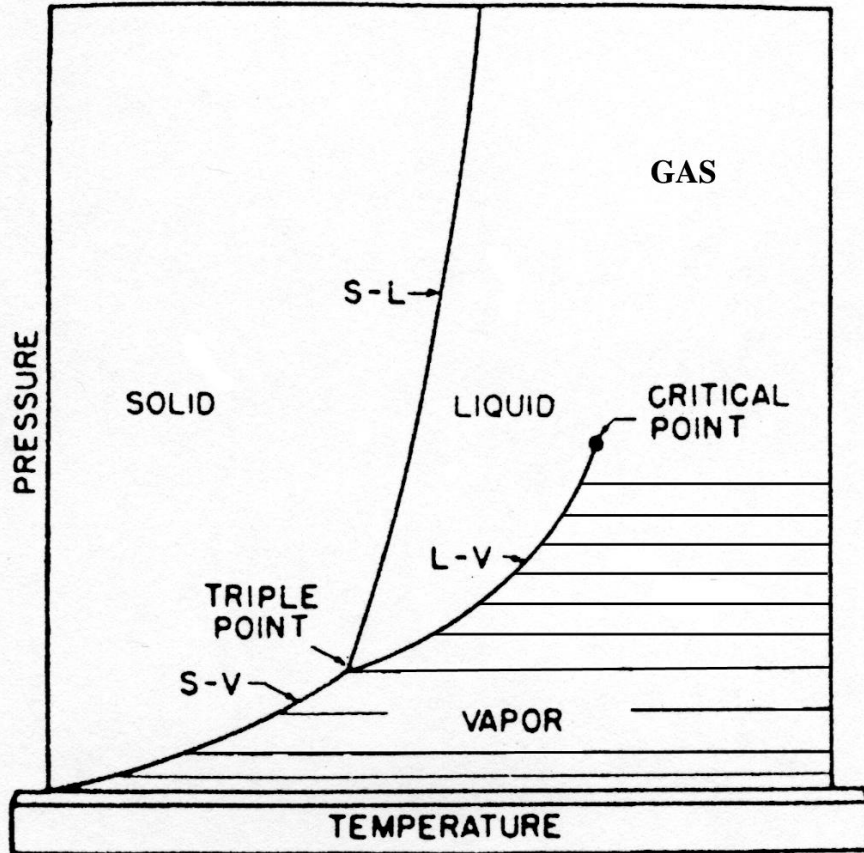
During b-c, the pressure is constant, which is called the saturation vapor pressure, or often just the vapor pressure (p_v).

At c, condensation is complete.

ΔV_{b-c} is the volume difference between the vapor and liquid phases at T_2 .

ΔV_{b-c} becomes smaller and vanishes at the critical point (p_c, V_c, T_c). Above T_c , the material cannot be condensed to a liquid, and it is called a gas; below T_c , it can be condensed and is called a vapor.

p - T phase diagram



The triple point is the only point where all three phases can exist simultaneously in equilibrium.

The vaporization of a material in the region of S-V below the triple point is called **sublimation**.

The vapor pressure curves (S-V and L-V) are of fundamental importance in controlling thin-film processes.

2.2 Maxwell-Boltzmann Distribution

Gas molecules are continuously colliding randomly with each other and with containing surfaces. The energy exchange taking place during these collisions leads to an equilibrium distribution of molecular speeds given by the Maxwell-Boltzmann formula:

$$\frac{dN/dv}{N} = 4\pi v^2 \left(\frac{m}{2\pi k_B T} \right)^{3/2} \exp \left(-\frac{\frac{1}{2}mv^2}{k_B T} \right)$$

N : total number of molecules in the system

v : molecular speed, m s^{-1}

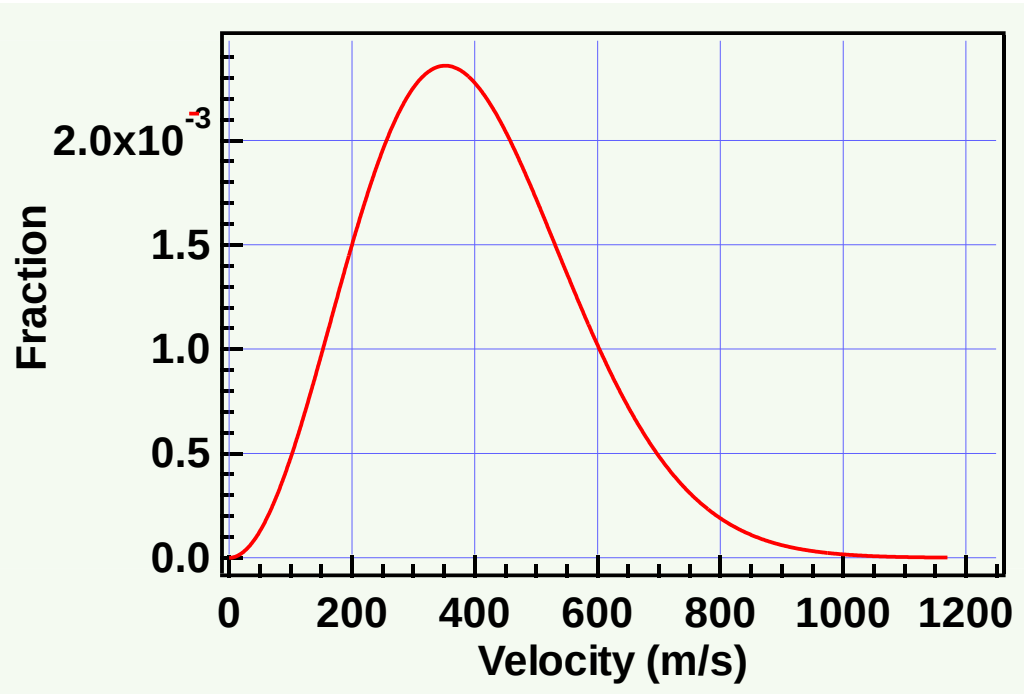
dN/dv : number of molecules dN within the speed increment dv at v

m : molecular mass, kg

k_B : Boltzmann's constant, $1.38 \times 10^{-23} \text{ J K}^{-1}$

T : absolute temperature, K

Maxwell-Boltzmann distribution for Ar at $T = 25\text{ }^{\circ}\text{C}$



$$\bar{v} = \sqrt{\frac{8k_{\text{B}}T}{\pi m}} = \sqrt{\frac{8RT}{\pi M}} \\ = 397\text{ m s}^{-1}$$

$$\sqrt{\overline{v^2}} = \sqrt{\frac{3k_{\text{B}}T}{m}} = \sqrt{\frac{3RT}{M}} \\ = 431\text{ m s}^{-1}$$

$\bar{v}$: mean speed

$\sqrt{\overline{v^2}}$: root-mean-square (rms) speed

$R = k_{\text{B}}N_{\text{A}}$: gas constant, $8.31\text{ J mol}^{-1}\text{ K}^{-1}$

N_{A} : Avogadro's number, $6.02 \times 10^{23}\text{ molecules mol}^{-1}$

$M = mN_{\text{A}}$: molar mass (molecular weight), kg mol^{-1}

2.3 Molecular Impingement Flux

The molecular impingement flux (molecules $\text{m}^{-2} \text{s}^{-1}$) is a fundamental determinant of the film deposition rate.

Let's consider a random distribution of molecular directions, that is, there is no bulk flow. This is always the case immediately above a surface, even if there is bulk flow farther away. For a given v_x , the corresponding molecular impingement flux would be given by

$$J_i = n v_x$$

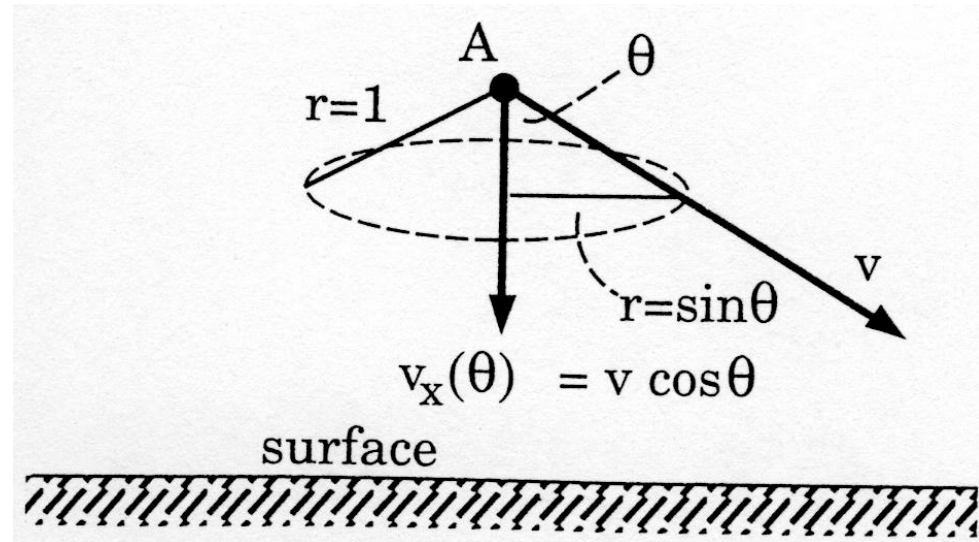
n : number density of molecules with v_x

$$\bar{v}_x = \frac{1}{2} v$$

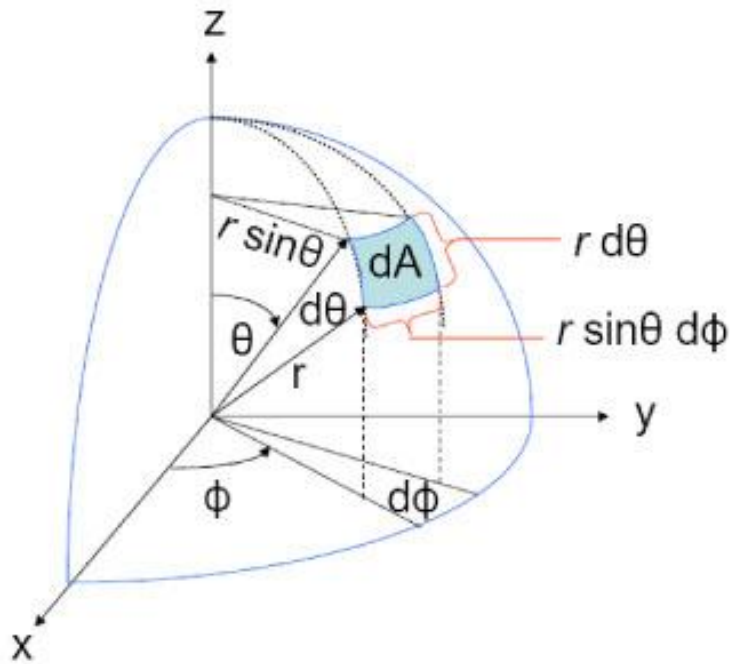
Because only a half of molecules move towards the surface, we have

$$J_i = \frac{1}{4} n v = \frac{1}{4} n \bar{v}$$

The meaning of n is also changed. It refers to the number density of all molecules.



Integrating over a Sphere

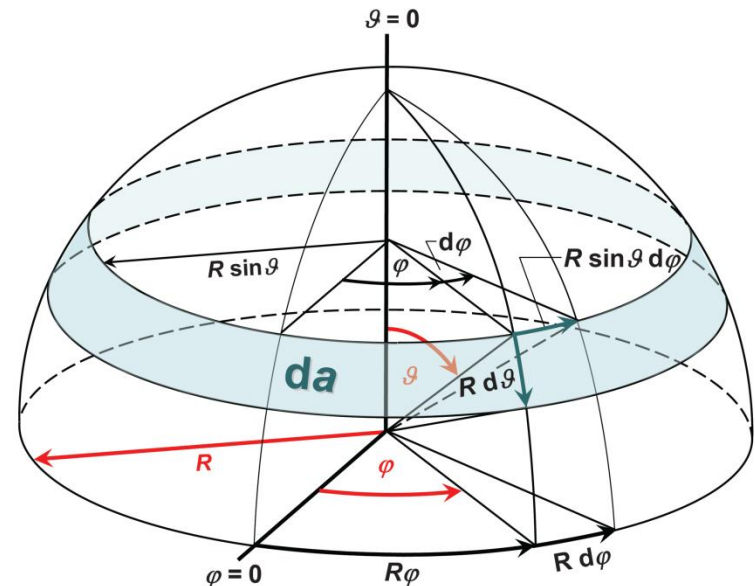


The shaded area

$$dA = r \sin \theta d\phi r d\theta = r^2 \sin \theta d\phi d\theta$$

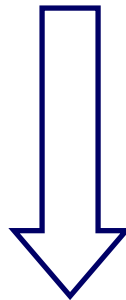
The shaded area

$$da = 2\pi R \sin \vartheta R d\vartheta = 2\pi R^2 \sin \vartheta d\vartheta$$



$$\frac{dN/dv}{N} = 4\pi v^2 \left(\frac{m}{2\pi k_B T} \right)^{3/2} \exp \left(-\frac{\frac{1}{2}mv^2}{k_B T} \right) \quad \text{Against } v$$

$$\frac{dn_x/dv_x}{n} = \left(\frac{m}{2\pi k_B T} \right)^{1/2} \exp \left(-\frac{\frac{1}{2}mv_x^2}{k_B T} \right) \quad \text{Against } v_x$$



$$J_i = \frac{1}{4} n \bar{v}$$

2.4 Ideal Gas Law

The ideal-gas law is an equation of state which expresses the relationship between pressure, volume, and temperature in the high-volume vapor and gas regions.

According to Newton, the force F exerted in a collision process between a molecule and a surface is equal to the rate of momentum change. For a surface area A ,

$$p = \frac{F}{A} = (2m\overline{v_x}) \frac{1}{2} (n\overline{v_x}) = nm\overline{v_x^2} = \frac{1}{3} nm\overline{v^2}$$

The equipartition theorem of classical statistical mechanics: energy is distributed equally among all degrees of freedom, and each one contains $(1/2)k_B T$ energy.

$$\varepsilon_t = \frac{1}{2} m\overline{v^2} = \frac{3}{2} k_B T$$

$$p = nk_{\text{B}}T = n_{\text{m}}RT = \frac{N_{\text{m}}RT}{V} = \frac{RT}{V_{\text{m}}}$$

p : pressure, N m^{-2}

$n = N/V$: molecular concentration, molecules m^{-3}

$n_{\text{m}} = n/N_{\text{A}}$: molar concentration, mol m^{-3}

$N_{\text{m}} = N/N_{\text{A}}$: number of moles of gas

V_{m} : molar volume, $\text{m}^3 \text{mol}^{-1}$

The ideal gas law is obeyed when the total volume of the molecules themselves in the gas is much smaller than the volume of the gas and when the cohesive forces between the molecules can be neglected.

2.5 Units of Pressure Measurement

Many different pressure units are frequently used in thin-film technology either for historical reasons or for convenience of size.

$$1 \text{ torr} = 133 \text{ Pa} = 1 \text{ mmHg}$$

$$1 \text{ psi} = 6894.757 \text{ Pa}$$

$$1 \text{ bar} = 750 \text{ torr} = 1.0 \times 10^5 \text{ Pa} = 0.99 \text{ standard atmosphere (atm)}$$

The standard conditions of T and p (stp) are 0°C and 1 atm (760 torr). At stp, the molecular density is $n = 2.69 \times 10^{19} \text{ molecules cm}^{-3}$. The molar volume is $V_m = 22400 \text{ cm}^3 \text{ mol}^{-1}$.

The term usually seen in gas supply monitoring is the “mass” flow rate measured in standard cm^3 per minute or standard cm^3 per second or standard liters per minute (sccm, sccs, slm). $1 \text{ sccm} = 4.48 \times 10^{17} \text{ molecules s}^{-1}$. In vacuum pumping, mass flow rate (pump throughput) is often measured in torr l s^{-1} or Pa l s^{-1} .

2.6 Knudsen Equation

The Knudsen equation is one of the most important relationships in thin-film technology.

$$J_i = \frac{N_A p}{\sqrt{2\pi MRT}}$$

Consider $M = 40 \text{ g mol}^{-1}$, $T = 25 \text{ }^\circ\text{C}$, $p = 0.001 \text{ Pa}$, $J_i = 2.4 \times 10^{15} \text{ molecules cm}^{-2} \text{ s}^{-1}$. For a molecular diameter of 0.3 nm , there are $1.4 \times 10^{15} \text{ molecules cm}^{-2}$ in a monolayer (ML, assume close packing and neglect empty interstitial areas). If all the molecules colliding on a surface stick on it in a thin-film deposition process, then the accumulation rate is $1.7 \text{ ML/s} = 1.8 \text{ }\mu\text{m/h}$.

This is a typical thin-film deposition rate. It also means that if this gas is a background impurity in the deposition chamber, the film is likely to incorporate a high level of contamination. To deposit a 99.9 % pure film at $1.8 \text{ }\mu\text{m/h}$, the p of impurity gases must be kept below $10^{-6} \text{ Pa} = 10^{-11} \text{ atm}$, which is considered to be in the range of ultrahigh vacuum (UHV).

The impingement flux is determined by the partial p of each gases. The film purity is determined by the ratio between the film molecular impingement flux and the impurity impingement flux. The film purity can be improved by decreasing the impurity partial p value.

In a thin-film process, the linear deposition rate, dh/dt , is easily determined from the film thickness, h , and the deposition time, t . The molecular deposition flux can be found from this rate:

$$J_i = \frac{dh}{dt} \left(\frac{\rho_m N_A}{M} \right)$$

Another quantity encountered in the study of gas adsorption is the **Langmuir**, L, which refers to the amount of exposure to a gas that a surface has seen. It is proportional to $p \cdot t$, and $1 \text{ L} = 10^{-6} \text{ torr} \cdot \text{s}$. If all impinging gas molecules adsorb, 1 L represents a surface accumulation of about 0.2 monolayer (?, considering the molecules mentioned on the last slide).

The Knudsen equation is very important because it relates the impingement flux to experimentally measurable quantities.